ARI DUTTA

Melbourne, VIC, Australia | +61 404 479 136 | aritro.dutta16@gmail.com
LinkedIn: linkedin.com/in/ar1tro | GitHub: github.com/echooff16 | Portfolio: echooff16.github.io/Portfolio
Work Rights: Full Working Rights in Australia

PROFESSIONAL SUMMARY

Software Engineering (Honours) graduate from **Monash University** who turns business problems into working applications. Delivered an **Azure-backed AI automation platform for 100+ users** at 98% uptime, automating document processing to cut manual effort by 95% and improving response accuracy by 40%. Experienced in gathering stakeholder requirements, building rapid prototypes, integrating systems via REST APIs, and designing relational SQL databases. **Available immediately in Melbourne.**

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript, TypeScript, React, FastAPI, SQL, PostgreSQL, MongoDB, PHP, HTML/CSS
AI & Machine Learning: PyTorch, YOLOv9, OpenCV, RAG Pipelines, LangChain, FAISS, LLM Integration, Prompt Engineering, Generative AI
Cloud & Infrastructure: Microsoft Azure (Azure AI, Blob Storage), REST APIs, OAuth 2.0, Supabase, Git, GitHub, CI/CD
Data Engineering & Automation: SQL, PostgreSQL, Python, Pandas, NumPy, Data Pipelines, ETL, Data Modelling, MongoDB, Automated Workflows
Development Tools: Claude, Cursor, GitHub Copilot, Gemini API, OpenAI API, Linux

RELEVANT EXPERIENCE

AI Developer | QEERI.AI *(Industry Placement)* Mar 2024 - Oct 2024 *Melbourne*

- Built AI Pathways, an AI platform that answers students' career questions from their assessment reports, using Python, Regex and PyMuPDF to extract report data into MongoDB, with an Azure AI-backed query layer.
- Improved response accuracy by **40%** by tracing out-of-scope answers through query logs, tightening prompt guardrails and validating changes with offline A/B testing.
- Built a full-stack web application using FastAPI, React and Azure AI Services, maintaining **98% uptime for 100+ active users**, with cloud storage on Azure Blob Storage.
- Built automated ETL and document processing pipelines across **500+ documents**, cutting **manual processing effort by 95%**.
- Worked with the founder and cross-functional stakeholders to translate commercial requirements into technical features, contributing to code reviews and iterative product updates and support.

Computer Vision Engineer | CSIRO *(Industry Placement)* Mar 2023 - Oct 2023 *Melbourne*

- Trained and evaluated custom **YOLOv9 object detection models in PyTorch**, achieving 92% mean average precision (mAP) across 50+ target categories for a government research deployment.
- Profiled and optimised the execution pipeline in Python, eliminating CPU/GPU bottlenecks to cut system latency by **15ms** for real-time edge processing.
- Diagnosed and resolved hardware/software integration issues through systematic root-cause analysis, **reducing hardware sync error rates below 1%**.
- Produced technical documentation and presented findings across interdisciplinary research and engineering teams.

Avionics Software Developer | Monash Uncrewed Aerial Systems Apr 2022 - Feb 2024 *Melbourne*

- Developed autonomous flight control software, contributing to a **15% increase in flight mission success rate**.
- Resolved communication and telemetry protocol errors across 10+ interconnected electronic systems, **reducing overall error rates by 25%**.
- **Platform achieved Top 10% global ranking at The New Flying Competition in Hamburg, Germany,** after I had moved on from the team in February 2024.

Freelance Web Developer | Creation Dec 2021 - Feb 2022 | 2024 - 2025

- **Designed, built and launched a full-stack e-commerce platform** using JavaScript, PHP, REST APIs and OAuth 2.0, with ongoing feature development and maintenance.

PROJECTS

AI Job Search Agent | Claude Code, Playwright MCP, Prompt Engineering 2026 - Present

- Designed an AI agent workflow that searches 50+ company careers pages through Greenhouse, Lever and Ashby job feeds, filters roles by eligibility rules and scores them against my resume.
- Built human-in-the-loop guardrails for form filling: field re-checks after resume parsing, stop on unexpected changes, and no automated submission.
- Logs ranked shortlists, search coverage and application status to CSV for weekly tracking.

Universal Book | React, TypeScript, Supabase, PostgreSQL, REST APIs *(Monash Final Year Project)* 2024

- Architected a real-time collaborative web application end-to-end, designing the relational PostgreSQL schema and custom RESTful API endpoints.

Logistics & Resource Optimisation Engine | Python, Graph Theory, Network Flow 2025

- Modelled locations and transport corridors as a directed weighted graph to identify sites reachable within a set distance limit
- Converted agent-to-site allocation into a multi-sink flow network, guaranteeing optimal assignment within capacity bounds

Music Pattern Analytics Engine | Python, Advanced Data Structures, String Algorithms 2025

- Built to identify and analyse repeated melodic patterns across songs, enabling efficient music similarity detection regardless of musical key.
- Indexed melodies using a custom suffix-tree structure for fast pattern lookup across all songs simultaneously
- Made search key-independent by comparing note intervals rather than absolute pitches, matching the same melody regardless of transposition

EDUCATION & CERTIFICATION

- **Bachelor of Software Engineering**, Monash University (DEC 2025)
- **HackerRank Software Engineer Certificate**
- **100 Days of Code**: Complete Python Pro Bootcamp, Udemy
- **ANZ Cyber Security Management Virtual Experience**

**References available on request*